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Ice Task 1

Big Data Analytics

Big Data Analytics refers to the processing and analysis of enormous datasets to find trends, patterns, and insights that contribute to organizations to make strategic business decisions. In Azure, it is facilitated by services such as the Azure Data Lake Storage (ADLS) to store large volumes of raw data, Azure Databricks to process and transform raw data with big data engines such as Apache Spark, and Azure Synapse Analytics to run sophisticated queries and create actionable data. Collectively, they make up the first part of the blue data pipeline: data is loaded into ADLS and processed and transformed in Databricks and analyzed in Synapse. They can be used in the analysis of customer behavior to optimize marketing strategies and to analyze large datasets in the healthcare sector to determine the trends of a disease and treatment results. (TechTarget, 2024) (GeeksforGeeks, 2025)

Real-Time Data Analysis

Real-Time Data Analysis is aimed at working with the data in real-time, when it is produced, which enables organizations to react instantly to the evolving circumstances. Azure, Azure Event Hubs receive real-time data streams sent by applications, devices or sensors and and Azure Stream Analytics processes real time streaming data and applies filters, aggregations and rules. The resulting processed information can be finally visualized using Power BI dashboards or be sent to alerting systems to act accordingly. This constitutes real-time portion of the Azure data pipeline, where insights are generated continuously, instead of in batches. Typical applications would be fraud detection in banking, where transactions are monitored in real time to prevent unauthorized transactions, and IoT-based industrial monitoring, where sensors notify in real time when equipment is malfunctioning or is underperforming. (Oracle, 2024) (TechTarget, 2024)

Integration in the Azure Data Pipeline

Big data and real-time analytics in Azure combined will offer a full-fledged data processing ecosystem. Organizations are able to accumulate and archive immense amounts of data to be analyzed in detail and at the same time track real-time data feeds to respond promptly to emergencies. When paired with such tools as ADLS, Databricks, Synapse, Event Hubs, and Stream Analytics, the Azure provides a flexible data pipeline capable of serving long-term strategic decision-making and operational decision-making in real-time. This two-pronged strategy enables the businesses to reduce value of their data by being proactive and responsive at the same time. (Microsoft Azure, 2025)

References

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